

GMM-Based Anomaly Detection: Learning Beyond Gaussian Assumptions

Suhani

24110358

IIT Gandhinagar
Gandhinagar, Gujarat, India

Mateen Sharief

24110197

IIT Gandhinagar
Gandhinagar, Gujarat, India

Mayank Choudhary

24110199

IIT Gandhinagar
Gandhinagar, Gujarat, India

Lingala Hasini

24110186

IIT Gandhinagar
Gandhinagar, Gujarat, India

Abstract

This project explores and extends the Gaussian Mixture Model (GMM)-based anomaly detection framework proposed by Zong et al. [14] (DAGMM). While DAGMM leverages deep autoencoders and GMMs to detect anomalies in a low-dimensional reconstruction space, it fundamentally relies on Gaussian distributional assumptions that may not hold for real-world data exhibiting heavy tails, skewness, or multimodality. We propose to systematically evaluate DAGMM’s limitations and extend it along three progressively expressive directions: (1) substituting Gaussian components with richer non-Gaussian mixture distributions—including Student- t , Laplace, and skew-normal; (2) replacing the parametric mixture entirely with a normalizing flow-based density estimator; and (3) combining both approaches in a hybrid Non-Gaussian Mixture Flow (NGMF) model. Additionally, we incorporate a Dirichlet Process (DP) mixture variant to remove the need for manual specification of the number of components. The goal is a more robust anomaly detection framework that retains the end-to-end trainability of DAGMM while generalizing well beyond Gaussian assumptions.

CCS Concepts

• **Computing methodologies** → **Anomaly detection**; *Density estimation*; *Deep learning*.

Keywords

anomaly detection, Gaussian mixture models, non-Gaussian distributions, deep autoencoder, normalizing flows, Dirichlet Process, density estimation

1 Introduction

Anomaly detection—identifying data points that deviate significantly from the majority—is a fundamental task in domains ranging from fraud detection and network intrusion to industrial fault diagnosis and healthcare monitoring. The challenge is amplified in high-dimensional settings, where the curse of dimensionality makes density estimation and threshold selection non-trivial.

The Deep Autoencoder Gaussian Mixture Model (DAGMM) [14] addresses this by combining a deep autoencoder with a GMM in a joint end-to-end framework. The autoencoder compresses high-dimensional inputs into a low-dimensional latent space, augmented with reconstruction-error features. A GMM is fit to this compressed

representation, and anomaly scores are derived from the estimated density—points in low-density regions are flagged as anomalies.

Despite its effectiveness, DAGMM makes a strong parametric assumption: that the data clusters in the latent space are Gaussian. Real-world anomaly detection datasets frequently violate this—distributions may be heavy-tailed, asymmetric, or otherwise non-Gaussian. This project proposes to:

- (1) Reproduce and rigorously benchmark DAGMM on standard anomaly detection datasets.
- (2) Identify failure modes attributable to Gaussian assumptions.
- (3) Develop and evaluate extensions using non-Gaussian mixture components, a Dirichlet Process mixture, and normalizing flow-based density models.

2 Related Work

DAGMM. Zong et al. [14] introduced DAGMM as an end-to-end deep learning framework for unsupervised anomaly detection. It jointly optimizes the autoencoder reconstruction loss and an energy-based loss derived from the GMM density estimate, using a sample energy function computed over the mixture components.

GMM limitations. Classical GMMs are well-studied but sensitive to outliers and ill-suited for heavy-tailed or skewed data [2]. The EM algorithm used to fit GMMs can degenerate under non-Gaussian noise, assigning high membership probabilities to anomalous points if they happen to lie near a component center.

Non-Gaussian mixtures. Student- t mixture models offer robustness to outliers via heavier tails controlled by the degrees-of-freedom parameter ν [10]. Skew-normal and skew- t distributions extend these to handle asymmetric clusters [7]. Laplace mixture models provide sparsity-inducing priors useful for high-dimensional settings.

Dirichlet Process mixtures. Nonparametric Bayesian approaches based on the Dirichlet Process [3] remove the need to fix the number of components K a priori. Tractable variational approximations via truncated stick-breaking have been integrated into neural network pipelines [9], making them suitable for end-to-end training.

Normalizing flows. Normalizing flows [11] learn an invertible transformation from a simple base distribution to an arbitrarily complex target, enabling exact likelihood computation. RealNVP [4]

and Glow [5] demonstrate their efficacy in high-dimensional density estimation, making them natural candidates for replacing the GMM component in DAGMM.

Other deep anomaly methods. Deep One-Class Classification (Deep SVDD [12]) and reconstruction-based approaches (VAE [6]) provide complementary baselines. Our focus, however, remains on density-based scoring within the DAGMM framework.

3 Methodology

We propose four progressively sophisticated components to improve upon DAGMM by relaxing Gaussian assumptions, along with a nonparametric extension. Section 3.1 establishes the baseline; Sections 3.2–3.4 introduce the three core strategies; Section 3.5 describes the Dirichlet Process variant.

3.1 Baseline: DAGMM Reproduction and Analysis

We first faithfully reproduce the DAGMM pipeline:

- (1) A compression network (deep autoencoder) encodes input $\mathbf{x}$ to latent vector $\mathbf{z}_c$, with reconstruction $\hat{\mathbf{x}}$. Relative Euclidean distance and cosine similarity between $\mathbf{x}$ and $\hat{\mathbf{x}}$ are concatenated to form $\mathbf{z} \in \mathbb{R}^d$.
- (2) An estimation network maps $\mathbf{z}$ to soft mixture assignments $\hat{\mathbf{y}} \in \mathbb{R}^K$ via softmax.
- (3) GMM parameters $\{\phi_k, \boldsymbol{\mu}_k, \Sigma_k\}_{k=1}^K$ are computed from the soft assignments, and sample energy $E(\mathbf{z})$ is defined as:

$$E(\mathbf{z}) = -\log \sum_{k=1}^K \phi_k \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_k, \Sigma_k). \quad (1)$$

- (4) The joint loss is:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda_1 \mathbb{E}[E(\mathbf{z})] + \lambda_2 \sum_k \|\text{diag}(\Sigma_k^{-1})\|_F. \quad (2)$$

We will profile DAGMM on datasets with known non-Gaussian structure to identify systematic failure modes before proposing extensions.

3.2 Strategy 1: Non-Gaussian Mixture Components

We replace the Gaussian mixture in DAGMM with richer distributional families, deriving appropriate energy functions for each.

Student- t Mixture Model (TMM). Each component follows a multivariate Student- t distribution with ν_k degrees of freedom:

$$p_k(\mathbf{z}) = \frac{\Gamma\left(\frac{\nu_k+d}{2}\right)}{\Gamma\left(\frac{\nu_k}{2}\right) (\nu_k \pi)^{d/2} |\Sigma_k|^{1/2}} \left(1 + \frac{\Delta_k(\mathbf{z})^2}{\nu_k}\right)^{-\frac{\nu_k+d}{2}}, \quad (3)$$

where $\Delta_k(\mathbf{z})^2 = (\mathbf{z} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1} (\mathbf{z} - \boldsymbol{\mu}_k)$. As $\nu_k \rightarrow \infty$, this recovers the Gaussian. The energy becomes $E_t(\mathbf{z}) = -\log \sum_k \phi_k p_k(\mathbf{z})$. The parameter ν_k is treated as learnable, providing a data-driven trade-off between Gaussian and heavy-tailed behaviour.

Laplace Mixture Model. Multivariate Laplace components provide an alternative heavy-tailed prior:

$$p_k(\mathbf{z}) \propto \exp\left(-\sqrt{\Delta_k(\mathbf{z})^2}\right), \quad (4)$$

which penalizes anomalies more aggressively in low-density regions. This is well-suited to intrusion detection or fraud datasets where anomalies are sparse and extreme.

Skew-Normal Mixture. For datasets with inherently asymmetric clusters, we incorporate skew-normal components [7] parameterized by location $\boldsymbol{\mu}_k$, scale Σ_k , and skewness vector $\boldsymbol{\alpha}_k$:

$$p_k(\mathbf{z}) = 2 \phi(\mathbf{z}; \boldsymbol{\mu}_k, \Sigma_k) \Phi\left(\boldsymbol{\alpha}_k^\top \Sigma_k^{-1/2} (\mathbf{z} - \boldsymbol{\mu}_k)\right), \quad (5)$$

where ϕ and Φ denote the Gaussian PDF and CDF, respectively. The soft membership assignments from the estimation network remain compatible across all three variants; only the energy computation changes.

3.3 Strategy 2: Normalizing Flow-Based Density Estimation

Our second strategy replaces the parametric mixture entirely with a normalizing flow, enabling the model to learn an arbitrarily complex density in the latent space. Given $\mathbf{z}$, we model its density via an invertible transformation $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$ (implemented as a RealNVP or masked autoregressive flow):

$$\log p(\mathbf{z}) = \log p_0(f_\theta(\mathbf{z})) + \log \left| \det \frac{\partial f_\theta}{\partial \mathbf{z}} \right|, \quad (6)$$

where p_0 is a simple base distribution (standard Gaussian or Laplace). The anomaly score is $-\log p(\mathbf{z})$. The joint training objective becomes:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} - \lambda_1 \mathbb{E}[\log p(\mathbf{z})], \quad (7)$$

where flow parameters θ and autoencoder parameters are optimized simultaneously. This approach makes no distributional assumptions about cluster shape and can capture multimodal, heavy-tailed, and skewed densities naturally.

3.4 Strategy 3: Hybrid Non-Gaussian Mixture Flow (NGMF)

Our third and most expressive approach combines the interpretability of mixture models with the flexibility of normalizing flows. We define a *Non-Gaussian Mixture Flow* (NGMF):

- (1) The estimation network assigns soft memberships to K components.
- (2) Each component k has its own lightweight normalizing flow f_{θ_k} , transforming $\mathbf{z}$ to a component-specific standardized space.
- (3) The per-component density is computed in that standardized space under a heavy-tailed base (e.g., Student- t).
- (4) The overall density is the mixture:

$$p(\mathbf{z}) = \sum_{k=1}^K \phi_k p_k(f_{\theta_k}(\mathbf{z})) \left| \det \frac{\partial f_{\theta_k}}{\partial \mathbf{z}} \right|. \quad (8)$$

This allows each cluster to have its own geometry, tailored non-Gaussian shape, and flexible transformation—potentially capturing complex data structure that neither a pure GMM nor a single flow can model.

3.5 Dirichlet Process Mixture Extension

Orthogonal to the three density strategies above, we also incorporate a Dirichlet Process mixture variant. Using a truncated stick-breaking approximation with a maximum of $K_{\max}$ components, the estimation network predicts stick-breaking logits from which mixture weights are derived via the softplus-stick-breaking transform [9]. A KL divergence term encourages unnecessary components to deactivate:

$$\mathcal{L}_{\text{KL}} = \text{KL}(q(\boldsymbol{\pi}) \parallel \text{DP}(\alpha_0, H)), \quad (9)$$

allowing the effective number of active components to be learned from data rather than specified as a hyperparameter. This variant can be applied on top of any of the three density heads above.

4 Datasets and Evaluation

Datasets. We evaluate on the following standard benchmarks:

- **KDD Cup 99** [1]: Network intrusion detection; tabular; ~494K samples; 41 features.
- **Thyroid / Arrhythmia** (UCI): Medical anomaly benchmarks with known non-Gaussian marginals.
- **CIFAR-10 / MNIST** (one-vs-rest): Image-level anomaly detection to test scalability to higher dimensions.
- **Synthetic heavy-tailed data**: Generated from Student- t and skew-normal mixtures to provide controlled non-Gaussian stress tests where ground-truth failure modes are known.

Metrics. We report Precision, Recall, F1-score, and AUROC at the best threshold, following [14]. All results are averaged over 20 independent runs to account for training variance.

Baselines. We compare against: original DAGMM [14], DAGMM-HO, Isolation Forest [8], Deep SVDD [12], One-Class SVM [13], and a standard VAE [6].

5 Progress Update Structure

Biweekly progress updates will address:

- (1) **Problem definition**: Refinement of the anomaly scoring formulation and distributional assumptions under investigation.
- (2) **Strategies**: Current experimental variants, ablations, and hyperparameter choices.
- (3) **Coding platform**: PyTorch for all deep learning components; scikit-learn for baselines; experiments tracked via Weights & Biases.
- (4) **Datasets**: Preprocessing status, train/test splits, and any data augmentation applied.

6 Timeline

- **Week 1 (Mar 27 – Apr 2)**: Environment setup, full literature review, and faithful DAGMM reproduction with baseline evaluation on KDD-99 and Thyroid.
- **Week 2 (Apr 3 – 9)**: Implement and evaluate Strategy 1 (Student- t , Laplace, and skew-normal mixture variants) and the DP extension; first biweekly update.

- **Week 3 (Apr 10 – 16)**: Implement Strategy 2 (normalizing flow density); begin Strategy 3 (NGMF) implementation; synthetic dataset experiments.
- **Week 4 (Apr 17 – 23)**: Finalize Strategy 3, comprehensive comparative evaluation across all datasets, ablation studies, and final report preparation; second biweekly update.

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